

PRACTICE QUESTIONS ON SERIALIZABILITY AND CONNCURRENCY

Question 1

Consider the following schedule involving three transactions. Identify all conflicting operations and construct the precedence graph. Hence determine whether the schedule is conflict serializable. If possible, write the equivalent serial schedule.

T1	T2	T3
R(X)		
	W(X)	
		R(Y)
W(Y)		
	R(Y)	
		W(X)

Question 2

Analyze the following schedule. Construct the serialization graph and determine whether the schedule is conflict serializable. Clearly justify your answer using the graph.

T1	T2
R(A)	
	W(A)
W(B)	
	R(B)
W(A)	

Question 3

Consider the following schedule:

T1	T2
R(A)	
	W(A)
W(A)	

Check whether the schedule is conflict serializable and view serializable. Explain the reason for the difference between the two results, if any.

Question 4

Consider the following locking schedule. Determine whether the schedule satisfies the Two Phase Locking protocol. Clearly identify the growing and shrinking phases of transaction T1 and comment on the possibility of cascading rollback.

T1	T2
LOCK-S(A)	
R(A)	
LOCK-X(B)	
W(B)	
UNLOCK(A)	
	LOCK-X(A)
UNLOCK(B)	
	W(A)

Question 5

A database system executes the following schedule. Analyze whether the above schedule follows Strict Two Phase Locking protocol. Also explain why Strict 2PL is preferred in practical database systems.

T1	T2
LOCK-X(X)	
W(X)	
	LOCK-S(X)
COMMIT	
UNLOCK(X)	
	R(X)

Question 6

Assume $TS(T1)=5$ and $TS(T2)=10$. Initially, $Read_TS(X)=0$ and $Write_TS(X)=0$.

Now consider the following schedule:

T1	T2
R(X)	
	W(X)
W(X)	

Apply timestamp ordering protocol on the above schedule and determine whether any transaction will be rolled back. Also update $Read_TS(X)$ and $Write_TS(X)$ after each operation.

Question 7

The following schedule is executed using timestamp ordering protocol. Given that $TS(T1)=5$ and $TS(T2)=10$, analyze each operation according to timestamp ordering rules and determine whether rollback occurs during execution.

T1	T2
	R(Y)
W(Y)	
	W(Y)

Question 8

Analyze the following schedule involving three transactions. Construct the wait-for graph and determine whether deadlock exists in the schedule. Also, identify which transaction should be rolled back to break the deadlock cycle.

T1	T2	T3
LOCK-X(A)		
	LOCK-X(B)	
		LOCK-X(C)
LOCK-X(B)		
	LOCK-X(C)	
		LOCK-X(A)

Question 9

Consider the following schedule. Analyze whether the schedule satisfies Strict Two Phase Locking protocol. Also comment on whether the schedule is recoverable and whether T2 can read uncommitted data.

T1	T2
LOCK-X(A)	

W(A)	
	LOCK-S(A)
COMMIT	
UNLOCK(A)	
	R(A)